

### **### Symptom: Firewall becomes completely unresponsive — no ping, no management access, no traffic passing**

#### **Overview**

A firewall that stops responding entirely (not just slow, but fully down) is a device-health emergency rather than a policy/configuration issue. The priority is determining whether this is a resource exhaustion problem (recoverable without a reboot) or a hardware/software fault (likely requiring one).

#### **Cause 1: CPU or Memory Exhaustion**

A firewall under sustained high CPU or memory load — from a traffic spike, a misbehaving process, or a memory leak in an aging software version — can become unresponsive to both management access and, in severe cases, data-plane traffic, even though the device is technically still powered on and running.

##### *Diagnostic steps:*

- If any management access remains (console, out-of-band interface), check current CPU and memory utilization immediately
- Review historical utilization graphs (if available) for a gradual climb versus a sudden spike, which points to different causes (memory leak vs. traffic event)
- Identify which process is consuming resources, if the platform exposes per-process statistics

#### **Cause 2: Disk Space Exhaustion**

Some firewall platforms become unresponsive or refuse new connections when local disk space (used for logging, session state, or temp files) fills completely — a less obvious cause than CPU/memory, but a common one on devices with verbose logging enabled and no log rotation configured.

##### *Diagnostic steps:*

- Check available disk space via console/out-of-band access if possible
  - Review logging configuration for excessive verbosity or missing rotation/retention policies that could allow logs to grow unbounded
  - Clear or archive old logs and diagnostic bundles as an immediate remediation step, separate from addressing the root logging configuration
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### **### Symptom: Firewall reboots unexpectedly, with no scheduled maintenance or manual action taken**

#### **Overview**

An unplanned reboot — the device comes back up on its own, sometimes with a gap in connectivity logged, sometimes discovered only via an uptime counter reset — needs root-causing before assuming it's a one-time fluke, since it can indicate a recurring hardware or software issue.

#### **Cause 1: Software Crash / Kernel Panic**

An unhandled software fault can cause the device's operating system to crash and automatically restart as a recovery mechanism. This is often logged in a crash dump or core file, if the platform generates one, and usually correlates with a specific software version's known bug rather than being random.

##### *Diagnostic steps:*

- Check for a generated crash dump or core file from the time of the reboot, and note the exact software version in use
- Search the vendor's known-issues/bug database for that software version and reboot symptom pattern before assuming it's environmental
- Review logs immediately preceding the reboot for an error pattern that consistently appears before each occurrence, if it's recurred more than once

#### **Cause 2: Power Event or Hardware Fault**

Not every unexpected reboot is software-caused — a brief power interruption, a failing power supply (on platforms with redundant PSUs, one failing can sometimes still cause a blip), or a hardware watchdog timer triggering due to a failing component can all produce the same symptom from the outside.

##### *Diagnostic steps:*

- Check environmental/power logs on the device itself and on any connected UPS or power monitoring equipment
- If the platform has redundant power supplies, confirm both are reporting healthy status, not just that the device is currently powered on
- Look for hardware fault codes or SEL (system event log) entries around the time of the reboot

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**### Symptom: Firewall is reachable and passing traffic, but management console/API access is extremely slow or times out**

## **Overview**

Unlike total unresponsiveness, this is a partial degradation — data plane (actual traffic forwarding) continues working while the control plane (management access) struggles. This split is a useful diagnostic signal on its own, since it usually rules out a full device failure.

### **Cause 1: Control Plane CPU Starved by Data Plane Processing**

On some firewall architectures, the same CPU resources are shared (or contended) between forwarding traffic and handling management requests. Under heavy traffic load, the device may prioritize keeping traffic flowing over responding to management sessions, causing exactly this "traffic fine, management slow" pattern.

#### *Diagnostic steps:*

- Check current traffic throughput/session count against the platform's rated capacity to see if it's operating near its limits
- Compare control-plane CPU (if reported separately from data-plane CPU on the platform) during the slow period
- Test management access during a lower-traffic period to see if responsiveness improves, which would support this cause

### **Cause 2: Too Many Concurrent Management Sessions**

Some platforms limit the number of simultaneous management sessions (SSH, API, GUI) and will queue or reject new connection attempts once that limit is reached, particularly if stale sessions from disconnected admins were never properly closed.

#### *Diagnostic steps:*

- Check the current count of active management sessions against the platform's documented limit
  - Review for stale or orphaned sessions (e.g., from a monitoring tool polling via API without properly closing connections) that could be consuming session slots unnecessarily
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### **### Symptom: High-availability (HA) pair shows both firewalls as active, or fails to fail over when the primary goes down**

#### **Overview**

In a redundant HA deployment, the whole point is a clean, automatic failover when the primary unit fails. When this doesn't happen correctly — either no failover occurs, or (worse) both units believe they're active simultaneously ("split-brain") — the redundancy design itself has broken down.

#### **Cause 1: Heartbeat/Sync Link Failure**

HA pairs typically communicate over a dedicated heartbeat or sync link to continuously confirm each other's status and replicate state. If this link fails (bad cable, misconfigured interface, switch port issue) while both units are otherwise healthy and reachable independently, each unit can lose visibility into the other's status and incorrectly assume the peer is down — potentially causing both to become active.

##### *Diagnostic steps:*

- Check the physical and logical status of the dedicated HA heartbeat/sync interface on both units
- Review HA state logs on both devices for when they last successfully communicated with each other
- Confirm the heartbeat link isn't sharing a path with general data traffic in a way that could cause it to be congested or accidentally disrupted

#### **Cause 2: HA State Synchronization Failure Without Link Failure**

Even with a healthy heartbeat link, HA units can fail to properly synchronize configuration or session state due to a version mismatch between the two units, a licensing discrepancy, or a sync process that silently stopped working — resulting in a failover that technically triggers but leaves the standby unit without correct, current state.

##### *Diagnostic steps:*

- Confirm both HA units are running identical (or HA-compatible) software versions and licensing
- Check HA sync status/logs for explicit synchronization errors, not just link-level connectivity
- Review the timestamp of the last successful full sync between the two units